

Visualization Design Rationale

Visualizing S&P 500 Volatility Regimes

DS4200: Information Presentation and Data Visualization

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Published website:

https://jeremy88bit.github.io/Visualizing_SP_500_Sector_Performance_Across_Volatility_Regimes/

Notes for the TA

All five visualizations run entirely in-browser. To view them, start a local server with `python -m http.server 8000` from inside the project folder and open `http://localhost:8000` in Chrome. There are no external dependencies to install. The site is also published at the GitHub Pages URL above.

Visualization 4 (Portfolio Strategy Comparison) was significantly upgraded beyond the original submission: the fixed 80%/40% allocation was replaced with two live sliders, a real-time Sharpe ratio display, and a max-Sharpe optimizer that grid-searches all 1% allocation combinations for each ETF. The standalone `viz4_portfolio_strategy.html` included in the ZIP reflects the original Python-generated version; `index.html` is the authoritative source for the current interactive implementation.

The cross-visualization brush link between Viz 1 and Viz 5 is one of the more distinctive interactions in the project. Dragging a date window on the Viz 1 overview panel automatically filters the return distribution histogram in Viz 5 to that period, connecting the timeline view of regime episodes directly to their distributional fingerprints.

Summary of Interaction Types

This project implements five distinct interaction types across the five visualizations:

1. Brush and zoom (Viz 1): drag selection on the overview panel updates the price and volatility panels in real time
 2. Hover tooltip (Viz 1, 2, 3, 4, 5): exact values on demand on every chart without cluttering the default view
 3. Click toggle (Viz 3): animated regime switching between All periods, Normal volatility, and High volatility correlation matrices with 500ms transitions
 4. Dropdown filter and live sliders (Viz 4, 5): sector and year selection, plus two allocation sliders in Viz 4 that update the chart and stats panel in real time
 5. Cross-visualization brush link (Viz 1 to Viz 5): dragging a date window in Viz 1 automatically filters the return distribution histogram in Viz 5 to that period
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ETF Color Palette

A fixed color identity is used for all six ETFs across every visualization:

- SPY (S&P 500): navy #1a5276
- XLK (Technology): purple #6c3483
- XLF (Financials): green #1e8449
- XLE (Energy): amber #b9770e
- XLV (Healthcare): red #c0392b
- XLY (Consumer Discretionary): blue #2874a6

Blue and red are reserved for regime encoding in Viz 3, 4, and 5, so no ETF color conflicts with the regime palette.

Visualization 1: Sector Volatility Timeline

Tool: Altair (Vega-Lite) | **Type:** Interactive: brush/zoom + tooltip + cross-viz link to Viz 5

Design Idea

A linked three-panel layout with a normalized price chart on top, rolling volatility in the middle, and a compact overview at the bottom. The overview serves as a brush selector: users drag to select a date range and both panels update to show that zoomed window. This "overview + detail" pattern follows Shneiderman's information-seeking mantra: overview first, zoom and filter, then details on demand.

Each ETF uses its fixed color identity across all five visualizations. The color legend is explicitly configured in the Altair spec with `orient: "right"` to ensure it renders reliably regardless of chart width, addressing instructor feedback that legends were missing from the line plots. Tooltips provide exact price, volatility, and regime label on hover. Prices are normalized to a common base (100) for fair cross-ETF comparison.

A cross-visualization link connects Viz 1 to Viz 5: brushing a date window automatically filters the return distribution histogram, letting users connect a market episode in the timeline to its distributional fingerprint.

Takeaway

Three distinct high-volatility regimes are visible: the 2018 Q4 rate-scare, the March 2020 COVID crash, and the 2022 rate-hike bear market. These episodes are not symmetric across sectors. During COVID, all six ETFs fell together but XLK recovered fully by end of 2020 while XLE was still down over 30%. In 2022 the pattern reversed: XLE gained roughly 60% while XLK fell more than 30%. The volatility subplot shows convergence during crises and divergence in calm periods.

Visualization 2: Volatility Regime Drivers

Tool: Altair (Vega-Lite) | **Type:** Static: hover tooltips with plain-English feature descriptions

Design Idea

The feature importance scores come from a [Random Forest classifier](#) trained to distinguish high- versus normal-volatility regimes. A Random Forest builds many decision trees on random subsets of the data and averages their predictions. Feature importance measures how much each variable reduces prediction error across all trees; higher importance means the model relied on that feature more to correctly classify which regime a day belongs to. This framing and the Wikipedia link were added in direct response to instructor feedback.

A horizontal bar chart is the most effective encoding for ranking. Bars are color-coded by feature category (Macroeconomic in red, Volatility in amber, Momentum in blue, Trend/MA in green, Range/Bands in purple, Volume in teal) so readers see which signal types dominate before reading individual labels. Hover tooltips show a plain-English description of each feature, addressing peer feedback that technical variable names were inaccessible. The chart is intentionally static: the message is a single ranked ordering.

Takeaway

The top two features, 20-day realized volatility (10.3%) and ATR ratio (9.7%), are direct measures of recent price turbulence. Price/200-day MA (8.5%) and 6-month momentum (7.3%) outrank shorter-term signals, suggesting regime shifts are associated with extended trend breakdowns rather than short-term noise. When FRED macro series are available, prior model versions showed Fed Funds Rate and VIX ranking in the top 3, indicating regime detection improves substantially when policy conditions are included.

Visualization 3: Sector Correlation Heatmap

Tool: D3.js | **Type:** Interactive: click toggle between regime states with 500ms animated transitions

Design Idea

A matrix heatmap lets viewers scan an entire $N \times N$ pairwise correlation space at a glance. The color scale is clamped to the actual data range (0.5 to 1.0) rather than the theoretical full range (-1.0 to 1.0). All sector correlations in this dataset sit above 0.50, so using the full range would compress all cells into indistinguishable deep red. Clamping to 0.5 to 1.0 spreads values across the full spectrum, directly addressing peer reviewer feedback. The gradient legend axis now correctly shows 0.5 at the bottom and 1.0 at the top.

Three toggle buttons switch between All periods, Normal volatility, and High volatility matrices. Clicking an already-active button (other than All periods) returns to All periods, addressing peer feedback that there was no way to deselect a toggle. Cell fills animate smoothly between states (500ms). Row and column labels use each ETF's dedicated color from the shared palette.

Takeaway

Across all periods, sector correlations are high: XLE sits at 0.82 to 0.85 with its peers, the lowest in the matrix, while XLK/XLF/XLY cluster above 0.88. Toggling to high-volatility periods shows correlations rising further across the board, compressing whatever diversification existed

in calm markets. Even XLE, structurally driven by commodity prices rather than rate expectations, provides limited independent movement during a market-wide shock.

Visualization 4: Portfolio Strategy Comparison

Tool: D3.js | **Type:** Interactive: live allocation sliders + dropdown + optimizer + Sharpe ratio

Design Idea

Cumulative return lines show both total performance and the path taken, including drawdowns and recoveries. The solid line is the tunable dynamic strategy; the dashed line is buy-and-hold. Using the same ETF color for both lines and differentiating by stroke style improves accessibility for colorblind readers.

Two live sliders replace the original fixed 80%/40% split: one for normal-regime equity allocation (50% to 100%) and one for high-volatility regime allocation (0% to 80%). Both sliders are continuous (no step constraint) and update the chart, statistics panel, and summary table in real time. A "Find max-Sharpe allocation" button runs a grid search across all 1% combinations to identify the optimal allocation for each ETF, displaying results in a clickable table where each row applies that ETF's optimal allocation to the sliders.

Strategy mechanics: hold the selected percentage in the sector ETF, remainder in cash (earning 0%), rebalancing daily at close. The stats panel shows total return, max drawdown, and annualized Sharpe ratio for both strategy and buy-and-hold. Sharpe is computed from actual daily percentage returns ($d.\text{returns} \times \text{allocation}$), not cumulative value differences, which was a bug in an earlier version that understated strategy Sharpe.

Takeaway

The default 80%/40% split trades meaningful total return for drawdown reduction. The sweet spot for most sectors sits around 90%/60%: a lighter touch that still reduces the worst drawdowns while giving up less upside. The "Find max-Sharpe" optimizer reveals the allocation that best balances return and volatility for each specific sector, and the corrected Sharpe calculation shows the strategy typically outperforms buy-and-hold on a risk-adjusted basis when the regime label has real predictive power.

Visualization 5: Return Distribution by Regime

Tool: D3.js | **Type:** Interactive: sector + year dropdowns, animated transitions, cross-viz date filter from Viz 1

Design Idea

Overlapping histograms with semi-transparent fills (opacity 0.5) compare two distributions in spread, center, and shape simultaneously. Blue represents normal volatility and red represents high volatility, matching the regime color language used throughout the site.

Two dropdowns (sector and year) enable cross-sectional and temporal comparisons. Animated bar-height transitions (500ms) prevent change blindness. A statistics panel shows exact mean, standard deviation, and observation counts for both regimes. The cross-visualization date filter

from Viz 1 lets users isolate the distributional fingerprint of a specific market episode without using the year dropdown.

Takeaway

For XLE across all years, the key difference between regimes is in spread rather than direction. Normal-regime std dev sits around 0.9%; high-volatility periods see this rise to roughly 1.7%, an increase of about 80%, not a doubling. Crucially, the means are close to zero in both regimes, meaning high-volatility periods are not reliably negative-return periods; they are high-variance periods. The roughly equal observation counts (~1,300 per regime) reflect the median-split design of the regime label. Individual years like 2020 show much more dramatic regime separation than the all-years average, because the all-period view averages across both crash and recovery episodes.

Data Design Notes

Feature Groups

The 28 engineered features fall into five groups:

- Market prices and returns: daily close, daily return, 5-, 20-, and 63-day cumulative price changes (per ETF)
- Volatility measures: 5-, 10-, 20-, and 50-day realized volatility (annualized), short/long volatility ratio, and term structure ratio
- Technical indicators: RSI (14-day), Bollinger Band width and position, ATR ratio, MA crossover signal, distances from 10-day and 50-day MAs, price-to-200-day MA, and 1-month, 3-month, and 6-month momentum
- Volume signals: 10-day and 50-day average volume and their ratio
- Macroeconomic conditions: VIX, Federal Funds Rate, and Unemployment Rate from FRED, forward-filled to daily frequency

Regime Label Definition

Each trading day is labeled "High" if its 20-day forward realized volatility exceeds the SPY median across the full dataset (approximately 14.8% annualized), and "Normal" otherwise. A median split produces a near-50/50 class balance by construction, which is why the observation counts in Viz 5 are always close to equal.

Preprocessing Steps

- Rolling window stabilization: first 200 rows per ETF dropped to allow 200-day MA calculations to produce valid values
 - FRED alignment: all three series forward-filled to daily trading-day frequency, then merged by date
 - Final dataset: 15,822 observations across 6 ETFs (approx. 2,637 per ETF), October 2015 through November 2025
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Response to Instructor Feedback

Instructor feedback on the Visualization Encoding criterion (27/30 pts):

"I assume the figure 2 comes from the random forest result. Include that in the interpretation and add Wikipedia link for the random forest will be better. Color legends are missing from the line plots."

Change 1: Random Forest explanation and Wikipedia link (Viz 2)

The Viz 2 description on the website now opens with a full explanation of what a Random Forest classifier is and why the importance scores come from it, directly framing the chart as a machine learning output rather than leaving readers to guess. A hyperlink to the [Wikipedia article on Random Forests](#) is embedded in the description. The takeaway paragraph was also updated to explicitly interpret the feature rankings as model-derived importance scores.

Change 2: Color legends added to line plots (Viz 1)

The Altair Vega-Lite specification for Viz 1 now explicitly configures the color legend with orient: "right", labelFontSize: 11, titleFontSize: 11, and symbolSize: 120. Previously the legend configuration was absent, leading to inconsistent rendering depending on chart width. The legend is now anchored to the right side of the main price panel and shows all six ETF ticker symbols with their corresponding colors. The volatility subplot below intentionally suppresses its legend to avoid duplication, which is standard multi-panel chart practice.

Response to Peer Review Feedback

Heatmap color scale shows no variation (changed)

Two reviewers independently noted that all heatmap cells appeared the same deep red because the color scale ran from -1.0 to 1.0 while all actual values sit between 0.50 and 1.0.

Change: the color scale is now clamped to 0.5 to 1.0. The gradient legend axis was also updated to show 0.5 at the bottom and 1.0 at the top, accurately reflecting the encoding.

S&P 500 and ETF missing from Key Terms (changed)

One reviewer noted the glossary omitted the most basic terms used throughout the site.

Change: "S&P 500," "ETF (Exchange-Traded Fund)," and "RSI (Relative Strength Index)" were added as the first three entries in the Key Terms glossary with plain-language definitions.

"What to look for" guides added (changed)

Multiple reviewers found the project dense and technical, making it hard to know what to focus on in each chart.

Change: a "What to look for" paragraph was added before each of the five visualizations, giving non-finance readers a one-sentence guide to the most meaningful pattern in each chart.

"What this means for investors" section added (changed)

Two reviewers asked for practical investor implications rather than only analytical conclusions.

Change: a new conclusion card titled "What this means for investors" was added covering three concrete takeaways: regime-aware sector allocation, the limits of the backtest (perfect foresight), and XLE as the most consistent but most volatile diversifier.

Strategy mechanics were unclear (changed)

Several reviewers could not follow the logic of the regime-aware strategy without knowing what it actually does.

Change: the Viz 4 description now opens with explicit mechanics (80% equities in normal regimes, 40% in high-vol, remainder in cash). The visualization was also upgraded from a fixed-split chart to a fully interactive optimizer with live allocation sliders, a real-time Sharpe ratio display, and a "Find max-Sharpe allocation" button that grid-searches all 1% combinations for each ETF.

Allocation sliders not continuous (changed)

The original sliders moved in 5% increments (step="5" attribute), which felt rigid when trying to fine-tune allocations.

Change: the step attribute was removed from both sliders, making them fully continuous. The optimizer grid search was also tightened from 5% to 1% steps to match the slider precision (approximately 4,100 combinations per ETF).

Sharpe ratio was incorrect (fixed)

The initial Sharpe calculation computed differences in cumulative portfolio values rather than actual daily percentage returns, producing understated strategy Sharpe ratios.

Fix: the Sharpe calculation now uses actual daily percentage returns (d.returns multiplied by the applicable allocation), which correctly shows the strategy outperforming buy-and-hold on a risk-adjusted basis when the regime label has predictive power.

Heatmap regime toggle could not be deselected (changed)

Multiple reviewers noted that clicking an already-active "Normal" or "High" button did nothing, with no way to return to "All periods" without refreshing.

Change: the toggle now uses a `_hmCurrentMode` state variable. Clicking an already-active non-All button returns to "All periods." Clicking "All periods" always stays on "All periods."

Navigation / tabs instead of scrolling (not changed)

Three reviewers suggested tabbed navigation or separate pages.

Decision: not implemented. The single-page layout is intentional: the five visualizations follow a narrative order where each chart answers a question raised by the previous one. Separating into tabs would break this continuity and sever the cross-visualization brush link between Viz 1 and Viz 5, which requires both charts in the same DOM. Section headers and numbered card backgrounds already provide orientation within the scroll.

Feature importance chart should be interactive (not changed)

One reviewer felt Viz 2 being static was inconsistent with the rest of the page.

Decision: not changed. The chart communicates a single ranked message. Adding filter interaction would distract rather than enhance. Hover tooltips with plain-English descriptions were added instead, which satisfies the spirit of the feedback without adding unnecessary interaction.